

```
#-----
# Vivado v2025.2 (64-bit)
# SW Build 6299465 on Fri Nov 14 19:35:11 GMT 2025
# IP Build 6300035 on Fri Nov 14 10:48:45 MST 2025
# SharedData Build 6298862 on Thu Nov 13 04:50:51 MST 2025
# Start of session at: Wed Jul 1 22:37:23 2026
# Process ID : 11808
# Current directory : C:/Users/ayode/Downloads/Parameterized
Matrix Multiply
Engine/Matrix_Accelerator/Matrix_Accelerator.runs/synth_1
# Command line : vivado.exe -log Matrix_Accelerator.vds
-product Vivado -mode batch -messageDb vivado.pb -notrace -source
Matrix_Accelerator.tcl
# Log file : C:/Users/ayode/Downloads/Parameterized
Matrix Multiply
Engine/Matrix_Accelerator/Matrix_Accelerator.runs/synth_1/Matrix_Ac
celerator.vds
# Journal file : C:/Users/ayode/Downloads/Parameterized
Matrix Multiply
Engine/Matrix_Accelerator/Matrix_Accelerator.runs/synth_1\vivado.jo
u
# Running On : AYODELE
# Platform : Windows Server 2016 or Windows 10
# Operating System : 26200
# Processor Detail : 13th Gen Intel(R) Core(TM) i5-1345U
# CPU Frequency : 2496 MHz
# CPU Physical cores : 10
# CPU Logical cores : 12
# Host memory : 16815 MB
# Swap memory : 1610 MB
# Total Virtual : 18426 MB
# Available Virtual : 4419 MB
#-----
source Matrix_Accelerator.tcl -notrace
Command: synth_design -top Matrix_Accelerator -part xc7k70tfbv676-1
Starting synth_design
Attempting to get a license for feature 'Synthesis' and/or device
```

```
'xc7k70t'
INFO: [Common 17-349] Got license for feature 'Synthesis' and/or
device 'xc7k70t'
INFO: [Device 21-403] Loading part xc7k70tfbv676-1
INFO: [Synth 8-7079] Multithreading enabled for synth_design using
a maximum of 2 processes.
INFO: [Synth 8-7078] Launching helper process for spawning children
vivado processes
INFO: [Synth 8-7075] Helper process launched with PID 22408
-----
Starting RTL Elaboration : Time (s): cpu = 00:00:03 ; elapsed =
00:00:04 . Memory (MB): peak = 1253.234 ; gain = 540.199
-----
INFO: [Synth 8-638] synthesizing module 'Matrix_Accelerator'
[C:/Users/ayode/Downloads/Parameterized Matrix Multiply
Engine/Matrix_Accelerator/Matrix_Accelerator.srcs/sources_1/new/Mat
rix_Accelerator.vhd:54]
INFO: [Synth 8-638] synthesizing module 'Systolic_Array'
[C:/Users/ayode/Downloads/Parameterized Matrix Multiply
Engine/Systolic_Array/Systolic_Array.srcs/sources_1/new/Systolic_Ar
ray.vhd:48]
    Parameter Rows bound to: 2 - type: integer
    Parameter Cols bound to: 2 - type: integer
    Parameter Input_Width bound to: 8 - type: integer
    Parameter Accumulator_Width bound to: 32 - type: integer
INFO: [Synth 8-638] synthesizing module 'Processing_Element'
[C:/Users/ayode/Downloads/Parameterized Matrix Multiply
Engine/Processing_Element/Processing_Element.srcs/sources_1/new/Pro
cessing_Element.vhd:48]
    Parameter Input_Width bound to: 8 - type: integer
    Parameter Accumulator_Width bound to: 32 - type: integer
INFO: [Synth 8-638] synthesizing module 'MAC_Unit'
[C:/Users/ayode/Downloads/Parameterized Matrix Multiply
Engine/MAC_Unit/MAC_Unit.srcs/sources_1/new/MAC_Unit.vhd:45]
    Parameter Input_Width bound to: 8 - type: integer
```

```
Parameter Accumulator_Width bound to: 32 - type: integer
INFO: [Synth 8-256] done synthesizing module 'MAC_Unit' (0#1)
[C:/Users/ayode/Downloads/Parameterized Matrix Multiply
Engine/MAC_Unit/MAC_Unit.srscs/sources_1/new/MAC_Unit.vhd:45]
INFO: [Synth 8-256] done synthesizing module 'Processing_Element'
(0#1) [C:/Users/ayode/Downloads/Parameterized Matrix Multiply
Engine/Processing_Element/Processing_Element.srscs/sources_1/new/Pro
cessing_Element.vhd:48]
INFO: [Synth 8-256] done synthesizing module 'Systolic_Array' (0#1)
[C:/Users/ayode/Downloads/Parameterized Matrix Multiply
Engine/Systolic_Array/Systolic_Array.srscs/sources_1/new/Systolic_Ar
ray.vhd:48]
INFO: [Synth 8-256] done synthesizing module 'Matrix_Accelerator'
(0#1) [C:/Users/ayode/Downloads/Parameterized Matrix Multiply
Engine/Matrix_Accelerator/Matrix_Accelerator.srscs/sources_1/new/Mat
rix_Accelerator.vhd:54]
WARNING: [Synth 8-6014] Unused sequential element
Gen_A_Row[0].Gen_A_Stage[1].Gen_A_Stage_N.A_Skew_reg[0][1] was
removed. [C:/Users/ayode/Downloads/Parameterized Matrix Multiply
Engine/Systolic_Array/Systolic_Array.srscs/sources_1/new/Systolic_Ar
ray.vhd:60]
WARNING: [Synth 8-6014] Unused sequential element
Gen_B_col[0].Gen_B_stage[1].Gen_B_Stage_N.B_Skew_reg[0][1] was
removed. [C:/Users/ayode/Downloads/Parameterized Matrix Multiply
Engine/Systolic_Array/Systolic_Array.srscs/sources_1/new/Systolic_Ar
ray.vhd:86]
-----
-----
Finished RTL Elaboration : Time (s): cpu = 00:00:05 ; elapsed =
00:00:05 . Memory (MB): peak = 1375.605 ; gain = 662.570
-----
-----
-----
-----
Start Handling Custom Attributes
-----
-----
```


Finished Handling Custom Attributes : Time (s): cpu = 00:00:05 ;
elapsed = 00:00:05 . Memory (MB): peak = 1375.605 ; gain = 662.570

Finished RTL Optimization Phase 1 : Time (s): cpu = 00:00:05 ;
elapsed = 00:00:05 . Memory (MB): peak = 1375.605 ; gain = 662.570

Netlist sorting complete. Time (s): cpu = 00:00:00 ; elapsed =
00:00:00.008 . Memory (MB): peak = 1375.605 ; gain = 0.000
INFO: [Project 1-570] Preparing netlist for logic optimization

Processing XDC Constraints

Initializing timing engine

Parsing XDC File [C:/Users/ayode/Downloads/Parameterized Matrix
Multiply

Engine/Matrix_Accelerator/Matrix_Accelerator.srcs/constrs_1/new/Clo
ckPeriod.xdc]

Finished Parsing XDC File [C:/Users/ayode/Downloads/Parameterized
Matrix Multiply

Engine/Matrix_Accelerator/Matrix_Accelerator.srcs/constrs_1/new/Clo
ckPeriod.xdc]

Completed Processing XDC Constraints

Netlist sorting complete. Time (s): cpu = 00:00:00 ; elapsed =
00:00:00 . Memory (MB): peak = 1455.184 ; gain = 0.000

INFO: [Project 1-111] Unisim Transformation Summary:

No Unisim elements were transformed.

Constraint Validation Runtime : Time (s): cpu = 00:00:00 ; elapsed
= 00:00:00.008 . Memory (MB): peak = 1455.184 ; gain = 0.000

Finished Constraint Validation : Time (s): cpu = 00:00:11 ; elapsed
= 00:00:12 . Memory (MB): peak = 1455.184 ; gain = 742.148

Start Loading Part and Timing Information

Loading part: xc7k70tfbv676-1

Finished Loading Part and Timing Information : Time (s): cpu =
00:00:11 ; elapsed = 00:00:12 . Memory (MB): peak = 1455.184 ; gain
= 742.148

Start Applying 'set_property' XDC Constraints

Finished applying 'set_property' XDC Constraints : Time (s): cpu =
00:00:11 ; elapsed = 00:00:12 . Memory (MB): peak = 1455.184 ; gain
= 742.148

INFO: [Synth 8-802] inferred FSM for state register 'State_reg' in
module 'Matrix_Accelerator'

State | New Encoding |
Previous Encoding

idle	00
00	
load	01
01	
drain	10
10	
valid	11
11	

INFO: [Synth 8-3354] encoded FSM with state register 'State_reg'
using encoding 'sequential' in module 'Matrix_Accelerator'

Finished RTL Optimization Phase 2 : Time (s): cpu = 00:00:11 ;
elapsed = 00:00:12 . Memory (MB): peak = 1455.184 ; gain = 742.148

Start RTL Component Statistics

Detailed RTL Component Info :

+---Adders :

2 Input	32 Bit	Adders := 1
---------	--------	-------------

+---Registers :

32 Bit	Registers := 9
8 Bit	Registers := 18

+---Muxes :

4 Input	32 Bit	Muxes := 1
2 Input	8 Bit	Muxes := 8
4 Input	2 Bit	Muxes := 1
2 Input	2 Bit	Muxes := 1
2 Input	1 Bit	Muxes := 7
4 Input	1 Bit	Muxes := 3

Finished RTL Component Statistics

Start Part Resource Summary

Part Resources:

DSPs: 240 (col length:80)

BRAMs: 270 (col length: RAMB18 80 RAMB36 40)

Finished Part Resource Summary

Start Cross Boundary and Area Optimization

DSP Report: Generating DSP

SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_Inst/MAC_Unit_Inst/Accumulator_Register_reg, operation Mode is:
(P+A*B) '.

DSP Report: register

SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_Inst/MAC_Unit_Inst/Accumulator_Register_reg is absorbed into DSP
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_Inst/MAC_Unit_Inst/Accumulator_Register_reg.

DSP Report: operator

SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_Inst/MAC_Unit_Inst/Accumulator_Register0 is absorbed into DSP
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_Inst/MAC_Unit_Inst/Accumulator_Register_reg.

DSP Report: operator

SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_Inst/MAC_Unit_Inst/Product is absorbed into DSP

SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_Inst/MAC_Unit_Inst/Accumulator_Register_reg.

DSP Report: Generating DSP

SA_Inst/Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.PE_Inst/MAC_Unit_Inst/Accumulator_Register_reg, operation Mode is: $(P+A2*B2)$ '.

DSP Report: register

SA_Inst/Gen_B_col[1].Gen_B_stage[1].Gen_B_Stage_N.B_Skew_reg[1][1] is absorbed into DSP

SA_Inst/Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.PE_Inst/MAC_Unit_Inst/Accumulator_Register_reg.

DSP Report: register

SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_Inst/A_Forward_reg is absorbed into DSP

SA_Inst/Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.PE_Inst/MAC_Unit_Inst/Accumulator_Register_reg.

DSP Report: register

SA_Inst/Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.PE_Inst/MAC_Unit_Inst/Accumulator_Register_reg is absorbed into DSP

SA_Inst/Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.PE_Inst/MAC_Unit_Inst/Accumulator_Register_reg.

DSP Report: operator

SA_Inst/Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.PE_Inst/MAC_Unit_Inst/Accumulator_Register0 is absorbed into DSP

SA_Inst/Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.PE_Inst/MAC_Unit_Inst/Accumulator_Register_reg.

DSP Report: operator

SA_Inst/Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.PE_Inst/MAC_Unit_Inst/Product is absorbed into DSP

SA_Inst/Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.PE_Inst/MAC_Unit_Inst/Accumulator_Register_reg.

DSP Report: Generating DSP

SA_Inst/Gen_PE_row[1].Gen_PE_col[0].Gen_Left_Edge.Gen_Left_Interior.PE_Inst/MAC_Unit_Inst/Accumulator_Register_reg, operation Mode is: $(P+A2*B2)$ '.

DSP Report: register
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_Inst/B_Forward_reg is absorbed into DSP
SA_Inst/Gen_PE_row[1].Gen_PE_col[0].Gen_Left_Edge.Gen_Left_Interior.PE_Inst/MAC_Unit_Inst/Accumulator_Register_reg.
DSP Report: register
SA_Inst/Gen_A_Row[1].Gen_A_Stage[1].Gen_A_Stage_N.A_Skew_reg[1][1] is absorbed into DSP
SA_Inst/Gen_PE_row[1].Gen_PE_col[0].Gen_Left_Edge.Gen_Left_Interior.PE_Inst/MAC_Unit_Inst/Accumulator_Register_reg.
DSP Report: register
SA_Inst/Gen_PE_row[1].Gen_PE_col[0].Gen_Left_Edge.Gen_Left_Interior.PE_Inst/MAC_Unit_Inst/Accumulator_Register_reg is absorbed into DSP
SA_Inst/Gen_PE_row[1].Gen_PE_col[0].Gen_Left_Edge.Gen_Left_Interior.PE_Inst/MAC_Unit_Inst/Accumulator_Register_reg.
DSP Report: operator
SA_Inst/Gen_PE_row[1].Gen_PE_col[0].Gen_Left_Edge.Gen_Left_Interior.PE_Inst/MAC_Unit_Inst/Accumulator_Register0 is absorbed into DSP
SA_Inst/Gen_PE_row[1].Gen_PE_col[0].Gen_Left_Edge.Gen_Left_Interior.PE_Inst/MAC_Unit_Inst/Accumulator_Register_reg.
DSP Report: operator
SA_Inst/Gen_PE_row[1].Gen_PE_col[0].Gen_Left_Edge.Gen_Left_Interior.PE_Inst/MAC_Unit_Inst/Product is absorbed into DSP
SA_Inst/Gen_PE_row[1].Gen_PE_col[0].Gen_Left_Edge.Gen_Left_Interior.PE_Inst/MAC_Unit_Inst/Accumulator_Register_reg.
DSP Report: Generating DSP
SA_Inst/Gen_PE_row[1].Gen_PE_col[1].Gen_Interior_Col.Gen_Interior.PE_Inst/MAC_Unit_Inst/Accumulator_Register_reg, operation Mode is:
(P+A''*B'')'.
DSP Report: register
SA_Inst/Gen_B_col[1].Gen_B_stage[1].Gen_B_Stage_N.B_Skew_reg[1][1] is absorbed into DSP
SA_Inst/Gen_PE_row[1].Gen_PE_col[1].Gen_Interior_Col.Gen_Interior.PE_Inst/MAC_Unit_Inst/Accumulator_Register_reg.
DSP Report: register
SA_Inst/Gen_A_Row[1].Gen_A_Stage[1].Gen_A_Stage_N.A_Skew_reg[1][1]

is absorbed into DSP
SA_Inst/Gen_PE_row[1].Gen_PE_col[1].Gen_Interior_Col.Gen_Interior.P
E_Inst/MAC_Unit_Inst/Accumulator_Register_reg.
DSP Report: register
SA_Inst/Gen_PE_row[1].Gen_PE_col[0].Gen_Left_Edge.Gen_Left_Interior
.PE_Inst/A_Forward_reg is absorbed into DSP
SA_Inst/Gen_PE_row[1].Gen_PE_col[1].Gen_Interior_Col.Gen_Interior.P
E_Inst/MAC_Unit_Inst/Accumulator_Register_reg.
DSP Report: register
SA_Inst/Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.P
E_Inst/B_Forward_reg is absorbed into DSP
SA_Inst/Gen_PE_row[1].Gen_PE_col[1].Gen_Interior_Col.Gen_Interior.P
E_Inst/MAC_Unit_Inst/Accumulator_Register_reg.
DSP Report: register
SA_Inst/Gen_PE_row[1].Gen_PE_col[1].Gen_Interior_Col.Gen_Interior.P
E_Inst/MAC_Unit_Inst/Accumulator_Register_reg is absorbed into DSP
SA_Inst/Gen_PE_row[1].Gen_PE_col[1].Gen_Interior_Col.Gen_Interior.P
E_Inst/MAC_Unit_Inst/Accumulator_Register_reg.
DSP Report: operator
SA_Inst/Gen_PE_row[1].Gen_PE_col[1].Gen_Interior_Col.Gen_Interior.P
E_Inst/MAC_Unit_Inst/Accumulator_Register0 is absorbed into DSP
SA_Inst/Gen_PE_row[1].Gen_PE_col[1].Gen_Interior_Col.Gen_Interior.P
E_Inst/MAC_Unit_Inst/Accumulator_Register_reg.
DSP Report: operator
SA_Inst/Gen_PE_row[1].Gen_PE_col[1].Gen_Interior_Col.Gen_Interior.P
E_Inst/MAC_Unit_Inst/Product is absorbed into DSP
SA_Inst/Gen_PE_row[1].Gen_PE_col[1].Gen_Interior_Col.Gen_Interior.P
E_Inst/MAC_Unit_Inst/Accumulator_Register_reg.

Finished Cross Boundary and Area Optimization : Time (s): cpu =
00:00:13 ; elapsed = 00:00:14 . Memory (MB): peak = 1455.184 ; gain
= 742.148

Sort Area is
SA_Inst/Gen_PE_row[1].Gen_PE_col[1].Gen_Interior_Col.Gen_Interior.P

-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
Note: The table above is a preliminary report that shows the DSPs
inferred at the current stage of the synthesis flow. Some DSP may
be reimplemented as non DSP primitives later in the synthesis flow.
Multiple instantiated DSPs are reported only once.

Finished ROM, RAM, DSP, Shift Register and Retiming Reporting

Start Applying XDC Timing Constraints

Finished Applying XDC Timing Constraints : Time (s): cpu = 00:00:19
; elapsed = 00:00:20 . Memory (MB): peak = 1613.492 ; gain =
900.457

Start Timing Optimization

Finished Timing Optimization : Time (s): cpu = 00:00:19 ; elapsed =
00:00:20 . Memory (MB): peak = 1638.328 ; gain = 925.293

Start Technology Mapping

Finished Technology Mapping : Time (s): cpu = 00:00:20 ; elapsed =
00:00:20 . Memory (MB): peak = 1638.328 ; gain = 925.293

Start IO Insertion

Start Flattening Before IO Insertion

Finished Flattening Before IO Insertion

Start Final Netlist Cleanup

Finished Final Netlist Cleanup

Finished IO Insertion : Time (s): cpu = 00:00:23 ; elapsed =
00:00:24 . Memory (MB): peak = 1875.871 ; gain = 1162.836

Start Renaming Generated Instances

Finished Renaming Generated Instances : Time (s): cpu = 00:00:23 ;
elapsed = 00:00:24 . Memory (MB): peak = 1875.871 ; gain = 1162.836

Start Rebuilding User Hierarchy

Finished Rebuilding User Hierarchy : Time (s): cpu = 00:00:23 ;
elapsed = 00:00:24 . Memory (MB): peak = 1875.871 ; gain = 1162.836

Start Renaming Generated Ports

Finished Renaming Generated Ports : Time (s): cpu = 00:00:23 ;
elapsed = 00:00:24 . Memory (MB): peak = 1875.871 ; gain = 1162.836


```
Start Handling Custom Attributes
-----
-----
-----
Finished Handling Custom Attributes : Time (s): cpu = 00:00:23 ;
elapsed = 00:00:24 . Memory (MB): peak = 1875.871 ; gain = 1162.836
-----
-----
-----
Start Renaming Generated Nets
-----
-----
-----
Finished Renaming Generated Nets : Time (s): cpu = 00:00:23 ;
elapsed = 00:00:24 . Memory (MB): peak = 1875.871 ; gain = 1162.836
-----
-----
-----
Start Writing Synthesis Report
-----
-----
-----
DSP Final Report (the ' indicates corresponding REG is set)
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+
|Module Name          | DSP Mapping  | A Size | B Size | C Size | D
Size | P Size | AREG | BREG | CREG | DREG | ADREG | MREG | PREG |
+-----+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+-----+
|MAC_Unit             | (P+A*B) '    | 30      | 18      | -       | -
| 32      | 0      | 0      | -       | -       | -       | 0      | 1      |
|Matrix_Accelerator  | (P+A'*B') '   | 30      | 18      | -       | -
| 32      | 1      | 1      | -       | -       | -       | 0      | 1      |
```

Matrix_Accelerator	(P+A'*B')'	30	18	-	-
32	1	1	-	-	-
0	1				
Matrix_Accelerator	(P+A''*B'')	30	18	-	-
32	2	2	-	-	-
0	1				

Report BlackBoxes:

BlackBox name	Instances

Report Cell Usage:

Cell	Count
1	1
2	8
3	4
6	6
7	11
8	7
9	33
10	36
11	226
12	25
13	129

Finished Writing Synthesis Report : Time (s): cpu = 00:00:23 ; elapsed = 00:00:24 . Memory (MB): peak = 1875.871 ; gain = 1162.836

Synthesis finished with 0 errors, 0 critical warnings and 0

warnings.

Synthesis Optimization Runtime : Time (s): cpu = 00:00:17 ; elapsed = 00:00:22 . Memory (MB): peak = 1875.871 ; gain = 1083.258

Synthesis Optimization Complete : Time (s): cpu = 00:00:23 ; elapsed = 00:00:24 . Memory (MB): peak = 1875.871 ; gain = 1162.836

INFO: [Project 1-571] Translating synthesized netlist

Netlist sorting complete. Time (s): cpu = 00:00:00 ; elapsed = 00:00:00.009 . Memory (MB): peak = 1885.039 ; gain = 0.000

INFO: [Netlist 29-17] Analyzing 12 Unisim elements for replacement

INFO: [Netlist 29-28] Unisim Transformation completed in 0 CPU seconds

INFO: [Project 1-570] Preparing netlist for logic optimization

INFO: [Opt 31-138] Pushed 0 inverter(s) to 0 load pin(s).

Netlist sorting complete. Time (s): cpu = 00:00:00 ; elapsed = 00:00:00 . Memory (MB): peak = 1887.707 ; gain = 0.000

INFO: [Project 1-111] Unisim Transformation Summary:

No Unisim elements were transformed.

Synth Design complete | Checksum: b86dc0ef

INFO: [Common 17-83] Releasing license: Synthesis

24 Infos, 2 Warnings, 0 Critical Warnings and 0 Errors encountered.

synth_design completed successfully

synth_design: Time (s): cpu = 00:00:27 ; elapsed = 00:00:29 .

Memory (MB): peak = 1887.707 ; gain = 1360.957

Write ShapeDB Complete: Time (s): cpu = 00:00:00 ; elapsed = 00:00:00.003 . Memory (MB): peak = 1887.707 ; gain = 0.000

INFO: [Common 17-1381] The checkpoint

'C:/Users/ayode/Downloads/Parameterized Matrix Multiply

Engine/Matrix_Accelerator/Matrix_Accelerator.runs/synth_1/Matrix_Accelerator.dcp' has been generated.

INFO: [Vivado 12-24828] Executing command : report_utilization -file Matrix_Accelerator_utilization_synth.rpt -pb

Matrix_Accelerator_utilization_synth.pb

INFO: [Common 17-206] Exiting Vivado at Wed Jul 1 22:37:59 2026...